

LINUX PROGRAMMING : ASSIGNMENT-2

1. What does the command pwd, whoami, and hostname display?

ANS The three fundamental Linux commands display core information about the user and the system's location:

- * pwd (print working directory) reveals the full, absolute path of the directory the user is currently operating in.
- * whoami outputs the login name of the user who is executing the command.
- * hostname shows the unique network name assigned to the computer or server.

2 Write the command to create a directory named "project" inside the /home/student folder and keep three .txt file into it. Give output snapshot

ANS mkdir /home/student/project

```
touch /home/student/project/file1.txt /home/student/project/file2.txt  
/home/student/project/file3.txt
```

3.Explain the difference between absolute path and relative path with proper examples

ANS Absolute Path

- * Always begins at the root directory (/).
- * It specifies the complete, fixed, and unique location of a file or directory within the entire file system.
- * The path remains the same regardless of the user's current working directory.
- * Example: /home/student/documents/notes.txt.
- * It is often preferred for use in scripts or commands where an unambiguous, specific location is required.

Relative Path

- * Starts from the current working directory (the directory the user is currently in).
- * The resulting location depends entirely on the user's starting point.
- * Uses special symbols like . (to represent the current directory) and .. (to represent the parent directory).
- * Example: If the current directory is /home/student/documents, the relative path to notes.txt is simply notes.txt or ./notes.txt. To access the student folder from documents, the relative path is ...
- * It is typically used for quick navigation within a local directory structure.

4What command will give you the already executed command traces in the terminal. Give output snapshot

ANS \$ history

```
42 ls -l  
43 cd /etc/  
44 grep "user" passwd
```

5. Compare the working functionality of find and locate command. Which one is faster and why?

ANS find Command

- * Operation: Searches the live filesystem in real-time.
- * Data Source: Scans the actual disk and directory structure starting from a specified path.
- * Accuracy: Highly accurate, as it checks the current, up-to-the-minute status of the filesystem.
- * Flexibility: Extremely flexible; allows for complex searches based on criteria like file name, size, modification time, permissions, and owner.
- * Speed: Slower because it requires accessing and reading the physical disk structure on the fly.

locate Command

- * Operation: Queries a pre-built database of file names and paths.
- * Data Source: Searches an indexed database (often updated daily by the updatedb utility).
- * Accuracy: May return outdated results if the file was created, moved, or deleted since the last database update.
- * Flexibility: Less flexible; primarily used for simple searches based only on the file name.
- * Speed: Faster because it queries an indexed database in memory rather than scanning the disk.

The locate command is faster because it searches an indexed database rather than scanning the entire filesystem in real-time.

6. Which command is used to modify file permissions in Linux? Give an example

ANS The command used to control file and directory access permissions in Linux is chmod (change mode).

Example:

Suppose you have a script named myscript.sh. To give the owner full read/write/execute permissions, while the group and others only have read and execute permissions, you would use the numeric (octal) mode:

- * Owner: rwx ($4+2+1 = 7$)
 - * Group: r-x ($4+0+1 = 5$)
 - * Others: r-x ($4+0+1 = 5$)
- chmod 755 myscript.sh

7. A file has permissions -rw -r- -r- -. What does this mean?

ANS The permission string -rw-r--r-- provides a breakdown of a file's type and the access rights granted to the owner, the group, and all others.

- * File Type (-): The initial hyphen indicates that the item is a regular file (not a directory, link, or special device).
- * Owner Permissions (rw-):

- * The Owner (User) has Read (r) and Write (w) permissions.
 - * The owner can view the file's contents and modify (change/delete) the file.
 - * They are denied execution permission (-).
 - * Group Permissions (r--):
 - * The Group that owns the file has Read (r) permission.
 - * Group members can view the file's contents but cannot modify or execute it.
 - * They are denied write (-) and execution (-) permissions.
 - * Others Permissions (r--):
 - * Others (everyone else on the system) have Read (r) permission.
 - * They can view the file's contents but cannot modify or execute it.
 - * They are denied write (-) and execution (-) permissions.
- In summary, only the file's owner has the authority to change the file's content.

8. Explain the difference between chown and chgrp with an example

ANS These two commands manage ownership, but on different levels:

- * chown (change owner): Alters the user who owns a file or directory. This is typically restricted to the root user for security reasons, though an owner can sometimes change a file's group.
- * Example: To make user alex the owner of a document:
- * chgrp (change group): Alters the group that owns a file or directory. A user must be a member of the new group to assign it to a file they own.
- * Example: To assign the file to the developers group:

9. A file needs to be accessible by multiple users but only writable by the owner. How will you set permissions?

ANS The requirement is that the file must be writable only by the owner and readable by everyone (owner, group, and others).

The needed permissions are:

- * Owner (User): Read and Write (\text{rw-})
- * Group: Read only (\text{r--})
- * Others (World): Read only (\text{r--})

`chmod 644 confidential_file.doc`

10. How do you check the manual page for any Linux commands?

ANS To access the official documentation or manual page (man page) for any Linux command, use the man command.

[8:59 pm, 27/9/2025] Pavan: `man [command_name]`

[8:59 pm, 27/9/2025] Pavan: `man cp`

[8:59 pm, 27/9/2025] Pavan: This opens a text-based viewer displaying the command's name, synopsis, detailed description, options, and exit status.